

Anti-Neurofilament light [MS-C2] Standard Size Ab00946-10.0

Isotype and Format: Human IgG1, Kappa

Clone Number: MS-C2

Alternative Name(s) of Target: PCDHGC3; NF-L; Neurofilament light polypeptide; 68 kDa neurofilament protein; Neurofilament triplet L protein; Protocadherin gamma-C3; Protocadherin gamma subfamily C 3, transcript variant 3

UniProt Accession Number of Target Protein: P07196; Q9UN70

Published Application(s): ELISA

Published Species Reactivity: Human

Immunogen: The original human IgG1 version of this antibody was obtained in the following manner: First, white matter tissue was dissected at autopsy from subjects with clinically defined MS, and individual B lineage cells were identified and captured. Multiplex PCR was used to amplify the Ig heavy and light chain variable regions, and recombinant IgGs were prepared from the Ig variable region sequences with IgG1 heavy chain and kappa constant domains.

Specificity: This antibody has been shown to bind to NF-L (neurofilament light) and protocadherin PCDHGC3.

Application Notes: The original human IgG1 version of the recombinant antibody MS-C2 was constructed in order to explore the antigen specificity of the Multiple Sclerosis B cell response (Willis et al, 2015; pmid: 26648933). The antibody was shown to bind two important neural antigens: neurofilament light (NF-L; strong binding with absorbance values that exceeded the mean +2SD of the control data set) and protocadherin PCDHGC3 (Willis et al, 2015; pmid: 26648933). Protocadherin gamma subfamily C, three (PCDHGC3) transcript variant three was of particular interest as a candidate autoantigen as protocadherin isoforms are predominantly expressed in the nervous system and have been implicated in human neurological disorders.

Antibody First Published in: Willis et al. Investigating the Antigen Specificity of Multiple Sclerosis Central Nervous System-Derived Immunoglobulins Front Immunol. 2015 Nov 25;6:600 [PMID:26648933](#)

Note on publication: Describes the application of recombinant antibodies from CNS lesion-derived B cells in antigen discovery in Multiple Sclerosis.

Product Form

Size:

200 µg Purified antibody.

Purification: Protein A affinity purified

Supplied In: PBS with 0.02% Proclin 300.

Storage Recommendation: Store at 4°C for up to 3 months. For longer storage, aliquot and store at -20°C.

Concentration: 1 mg/ml.

Important note – This product is for research use only. It is not intended for use in therapeutic or diagnostic procedures for humans or animals.